



ĐẠI HỌC
BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY
OF SCIENCE AND TECHNOLOGY

SEMANTIC WEB • TOPIC 2

Vietnamese DBpedia

Building a DBpedia version
for the Vietnamese language

Group 23

Ha Thi Kim Phung 20242460M

Do Ngoc Trung 20232311M

Bui Duc Phan Hoang 20252004M

Do Hoang Dung 20241184M

Lecturer: Dr. Do Ba Lam

ONE LOVE. ONE FUTURE.

The Assignment and What We Built

Task	What we built	Result
1. Define an ontology	<code>vio:</code> in OWL, every class under a <code>dbo:</code> class	18 classes, 54 properties
2. Collect articles	entities chosen via Wikidata, articles via the MediaWiki API	990 articles
3. 4-star data	infobox extraction, Unicode IRIs, quality checks, OWL 2 RL	131,543 triples
4. Link to DBpedia	<code>owl:sameAs</code> from the Wikidata sitelink of the same item	777 links
5. SPARQL endpoint / terminal	<code>/sparql</code> endpoint, query CLI, Linked Data, web UI	endpoint, CLI, web UI

Domain: in depth rather than in breadth

Vietnamese **football** (players, clubs, national teams, stadiums), **provinces** and **universities** — not all of Vietnamese Wikipedia

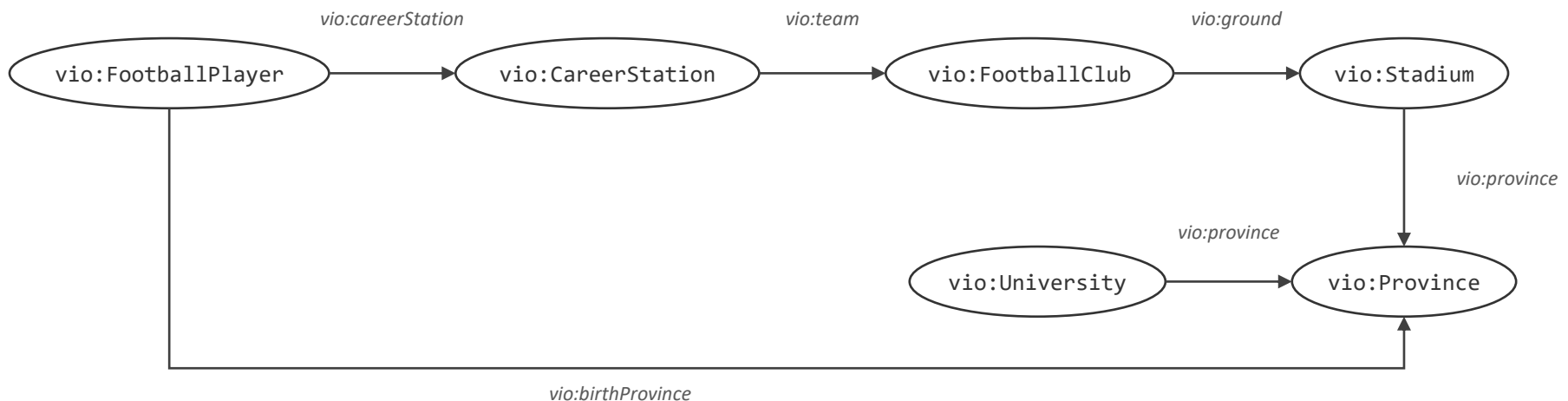
Why This Domain?

990 entities

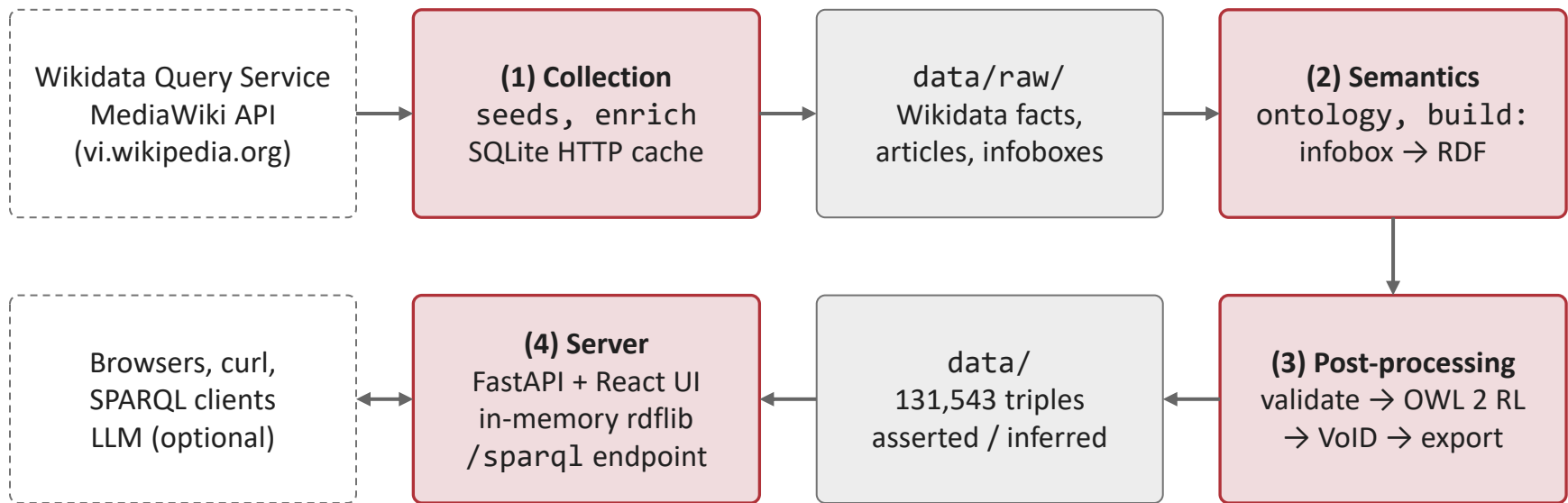
- 611 football players
- 61 clubs, 10 national teams
- 29 stadiums
- 168 universities
- 110 provinces (34 current)
- 1 country

Chosen because they are

- **densely linked** → multi-hop questions
- rich in infoboxes: **97%** of articles
- linkable: **78%** have an English article
- selected through Wikidata (P31, country Q881), not through viwiki categories, which editors assign inconsistently



Architecture: Four Layers, Seven Commands



```
python -m vidbpedia <step>
```

```
seeds → enrich → ontology → build → postprocess --strict → serve  
then query from the terminal · React UI: python -m ui.api
```

- Only layer (1) calls the network; raw downloads are committed → the dataset rebuilds **offline**, identically
- Graphs built with the rdflib API; **one function** creates every IRI

Collecting Entities and Articles

Wikidata Query Service — one query per class, e.g. players:

```
?item wdt:P31 wd:Q5 ;                                # human
      wdt:P106 wd:Q937857 ;                            # association football player
      wdt:P27 wd:Q881 .                                # citizen of Vietnam
?vs schema:about ?item ; schema:name ?viTitle ;
   schema:isPartOf <https://vi.wikipedia.org/> .      # has a Vietnamese article
```

- drop list and disambiguation pages; English article optional
- facts in blocks of 300: dates **with precision**, quantities with point in time, relations with start/end/apps/goals qualifiers
- birthplaces and locations mapped to a province through P131*

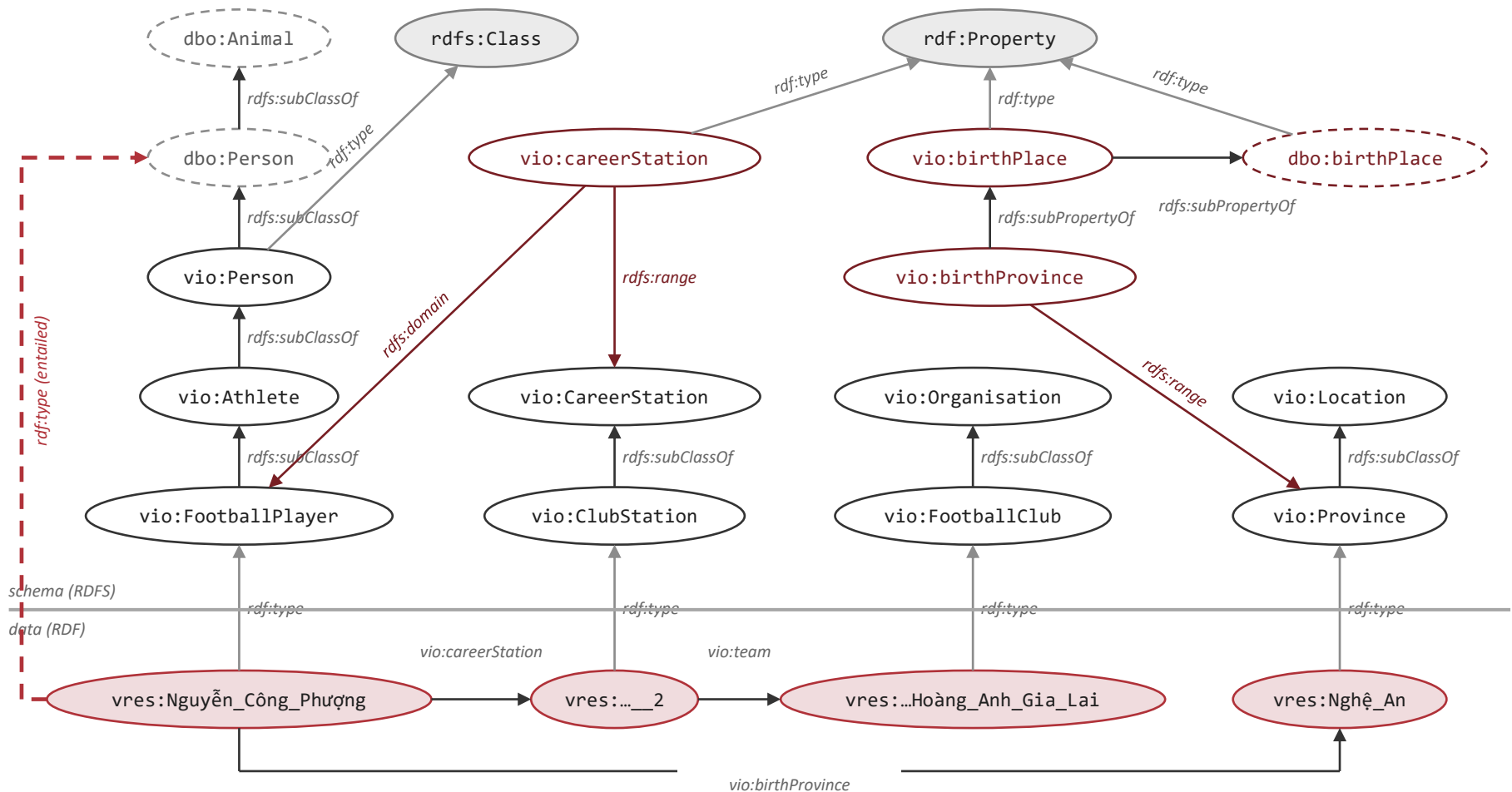
MediaWiki API (vi.wikipedia.org), batches of up to 50 titles

- page ID, revision, image, coordinates, lead section (abstract)
- categories, redirects, infobox parameters, external links
- 2,588 infobox link targets resolved (71 disambiguation pages)

Polite and reproducible

2 s / 1 s between requests, maxlag, retries on 429 · SQLite cache, --offline and --refresh

Ontology vio: as an RDFS Graph



Excerpt: 18 classes and 54 properties in total, every class $\sqsubseteq$ a `dbo: class` · dashed: `rdfs:type` `dbo:Person` entailed by RDFS

OWL Axioms Do the Work

Axiom	Definition	After reasoning
Property chain	$\text{careerStation} \circ \text{team} \sqsubseteq \text{playedFor}$	2,772 player–team edges
Inverse	$\text{hasPlayer} \equiv \text{playedFor}^{-}$	clubs list their players
$\exists$ restriction	$\text{FootballPlayer} \sqcap \exists \text{careerStation} . \text{NationalTeamStation} \sqsubseteq \text{NationalTeamPlayer}$	486 national-team players
$\forall$ restriction	$\text{ClubStation} \sqsubseteq \forall \text{team} . \text{FootballClub}$	154 more clubs typed
Alignment	every <code>vio:</code> class, 38 properties $\sqsubseteq$ <code>dbo:</code>	<code>dbo:Person</code> : 0 $\rightarrow$ 659
Disjointness	4 <code>owl:AllDisjointClasses</code>	caught 2 modelling bugs

Designed from competency questions and the course's modelling rules

“Which clubs has a player played for?” · “How many provinces are there now?” · “Which clubs play at home in Hà Nội?” — each is now an automated test

Two layers: reuse `dbo:` with Vietnamese labels; a `vio:` class only where an axiom needs it (no `vio:Animal`); `vio:` $\sqsubseteq$ `dbo:`, never $\equiv$

From Wikitext to RDF

Infobox extraction

1. **Render** nested templates (dates, heights, lists); drop refs
2. **Parse** Vietnamese values: 16.361, 2, 2015–, → (mượn) (on loan)
3. **Map** keys to `vio:` for 5 template families
4. **Choose** Wikidata or infobox per property

Career rows `yearsn/clubs/caps/goalsn`
→ 3,382 CareerStations for 601 players

DBpedia datasets reproduced

- labels (vi, en), abstracts: 990
- page IDs, revisions, provenance
- images: 543
- categories: 9,126 links
- redirects: 1,274
- external links, homepages, coordinates
- raw infobox `vip::` 24,605 triples

IRIs keep Unicode, like DBpedia chapters

`http://vi.dbpedia.org/resource/Nguyễn_Công_Phượng` · stations ...__1, ...__2, ... ·
redirects → `skos:altLabel`

Transform to 4★: One Infobox, Before and After

Before: wikitext — open and machine-readable, but text that names nothing with a URI (≤ 3 ★)

```
| birth_date   = {{ngày sinh và tuổi|1995|1|21}}  
| birth_place = [[Đô Lương (huyện)|Đô Lương]], [[Nghệ An]], [[Việt Nam]]  
| height      = {{convert|1,69|m}}  
| years2 = 2016 | clubs2 = → [[Mito HollyHock]] (mượn) | caps2 = 5 | goals2 = 0
```

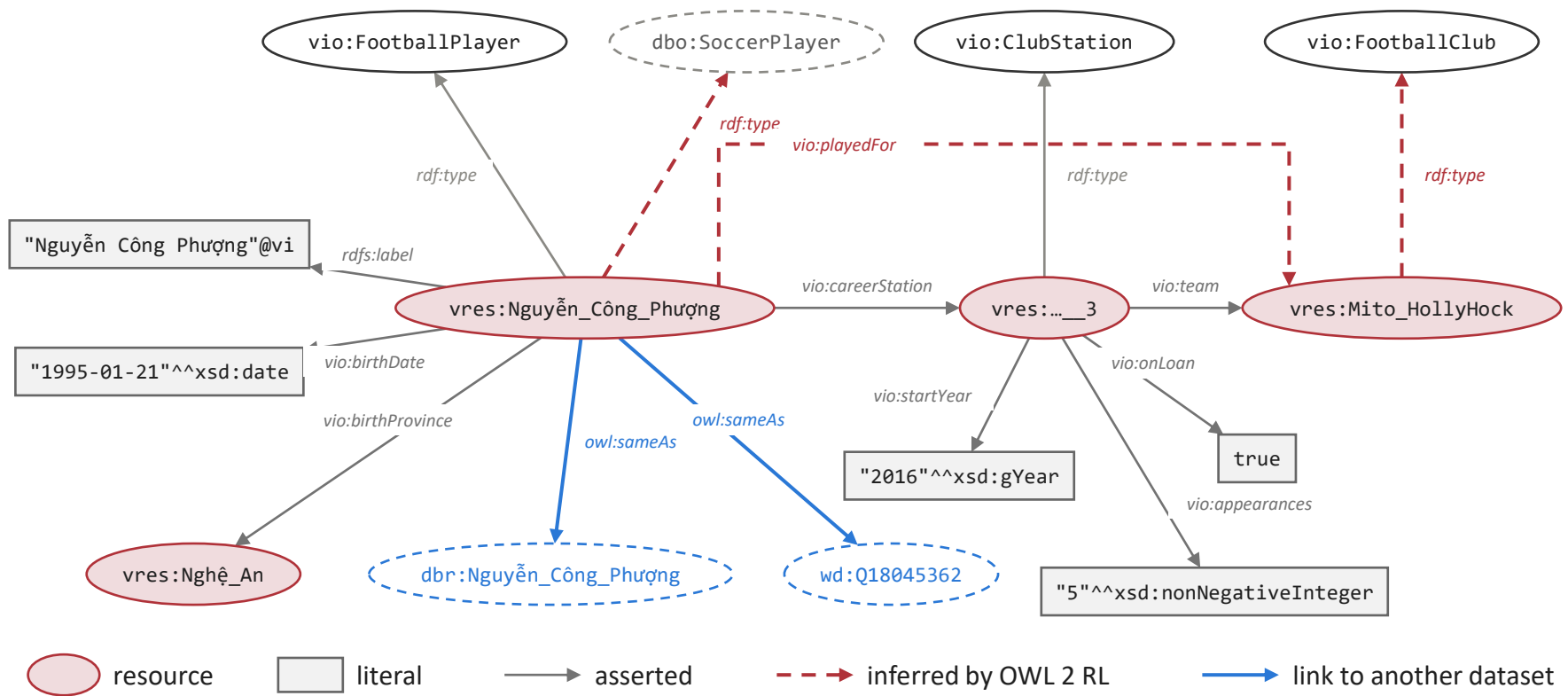
↓ python -m vidbpedia build

After: RDF with an HTTP IRI for every thing (4★)

```
vres:Nguyễn_Công_Phượng a vio:FootballPlayer ;  
    vio:birthDate "1995-01-21"^^xsd:date ;  
    vio:birthPlace vres:Đô_Lương_(huyện) ;    vio:birthProvince vres:Nghệ_An ;  
    vio:height 1.68e+00 ;                      # xsd:double; Wikidata wins over 1,69  
    vio:careerStation vres:Nguyễn_Công_Phượng__3 .  
vres:Nguyễn_Công_Phượng__3 a vio:ClubStation ;  
    vio:team vres:Mito_HollyHock ;    vio:onLoan true ;  
    vio:startYear "2016"^^xsd:gYear ;    vio:endYear "2016"^^xsd:gYear ;  
    vio:appearances "5"^^xsd:nonNegativeInteger ;  
    vio:goals "0"^^xsd:nonNegativeInteger .
```

IRIs for the player, places, clubs and the loan spell itself · **typed literals** (xsd:date, gYear, nonNegativeInteger, boolean) · links and templates **interpreted** · **shared vocabularies** (vio: $\sqsubseteq$ dbo:)

One Entity, End to End



1 of the player's 11 career stations (the 2016 loan of slide 9) · lecture notation (02_RDF): ellipse = resource, rectangle = literal

Validation and OWL 2 RL Reasoning

8 checks before export

TBox (kinds, declared terms, dbo:
ancestors, labels, isDefinedBy, names) ·
object vs. datatype values · literal ranges ·
required fields · functional properties ·
object ranges · owl:sameAs targets ·
plausible values

0 errors, 0 warnings

152 tests pass

What reasoning adds (+41,730 triples, 35 s)

Query	Asserted	Published
?x a dbo:Person	0	659
?x a dbo:SoccerPlayer	0	611
?x a dbo:Place	0	370
?x vio:playedFor ?t	0	2,772
?t vio:hasPlayer ?x	0	2,772
?x dbo:team ?t	0	3,699

Input: types and vio: triples only; no sameAs, functional or
cardinality axioms

Reasoning caught two modelling bugs

- (1) National teams inferred as *clubs*, a disjointness violation → domain of ground/manager changed to Organisation.
- (2) “Clubs at home in Hà Nội” returned clubs from Đà Nẵng → removed tenant $\equiv$ ground⁻.

Linking to English DBpedia (5★)

How

- Wikidata item → its English sitelink → owl:sameAs dbr:<title>
- no matching by label, which is ambiguous for Vietnamese names
- all 990 entities also → wd:Q...
- validation: no list/disambiguation targets, no shared targets
- described as two VoID linksets

Class	Linked / total
Players	508 / 611
Clubs, national teams	70 / 71
Stadiums	25 / 29
Universities	89 / 168
Provinces, country	85 / 111
Total	777 / 990

Checked on the live DBpedia

dbr: target	Links	Share
Article	733	94.3%
Redirect	12	1.5%
Not in DBpedia	32	4.1%

Our titles come from today's English Wikipedia; DBpedia is an older snapshot (16 of the 32 missing targets are clubs)

Confirms DBpedia uses the same Unicode IRIs, e.g. dbr:Đặng_Văn_Lâm

Dereferenceable IRIs and the SPARQL Endpoint

Route	Response
/resource/{name}	303 → HTML page, or → /data/{name}.ext when Accept asks for RDF
/page/{name}	303 → HTML page, like dbpedia.org/page/...
/data/{name}.ext	triples about the resource: .ttl, .nt, .jsonld, .rdf
/ontology/{term}	definition of a <code>vio:</code> term, in Turtle
/sparql	SPARQL 1.1 Protocol: GET <code>?query=</code> or POST; JSON, XML, CSV or RDF by Accept; read-only

Recorded on the running server:

```
$ curl -si -H "Accept: text/turtle" \  
    http://127.0.0.1:8000/resource/Nguy%E1%BB%85n_C%C3%B4ng_Ph%C6%B0%E1%BB%A3ng  
HTTP/1.1 303 See Other  
vary: Accept  
location: /data/Nguy%E1%BB%85n_C%C3%B4ng_Ph%C6%B0%E1%BB%A3ng.ttl  
$ curl -sL -H "Accept: text/turtle" http://127.0.0.1:8000/resource/Nguy%E1%BB%85n_  
vres:Nguyễn_Công_Phượng a dbo:Animal, dbo:Athlete, dbo:Person, dbo:SoccerPlayer, ... ;  
    dbo:birthDate "1995-01-21"^^xsd:date ;  
    ...  
    vio:birthProvince vres:Nghệ_An ;  
    ...  
    owl:sameAs dbr:Nguyễn_Công_Phượng, wd:Q18045362 ;
```

Headers and Turtle abridged. Browsers go to `?resource=<name>`, which opens the entity page `/entity/<name>`. Clients must percent-encode the Vietnamese characters, as browsers do.

Five Stars, Checked

Star	Evidence
★ open licence	CC BY-SA 4.0 in the VoID description and the ontology; dumps in a Git repository
★★ machine-readable	RDF triples, not text or tables
★★★ open formats	Turtle and N-Triples dumps; each resource also as JSON-LD and RDF/XML
★★★★ URIs	an HTTP IRI for every entity, station, category and redirect; dereferenced with 303 + content negotiation
★★★★★ linked	777 owl:sameAs to DBpedia (733 verified), 990 to Wikidata; classes and properties $\sqsubseteq$ dbo:

Caveat on the fourth star

The `vi.dbpedia.org` namespace follows the DBpedia-chapter convention, but we do not own that domain: the IRIs resolve only through our running server (`http://127.0.0.1:8000/resource/...`).

Web Interface: Five Pages

Nguyễn Công Phượng
Thực thể thuộc lớp **Cầu thủ bóng đá** - đồ thị <http://vi.dbpedia.org>
http://vi.dbpedia.org/resource/Nguyễn_Công_Phượng

[Wikipedia](#) [DBpedia EN](#) [Wikidata Q18045362](#) [Linked Data \(Turtle\)](#)

122 triple khai báo · **38 triple suy luận**

Cây phân lớp

Lớp khai báo và các lớp DBpedia suy ra bằng OWL 2 RL (rdfs:subClassOf).

- [dbo:Animal](#) [suy luận](#)
- [dbo:Person](#) [suy luận](#)
- [dbo:Athlete](#) [suy luận](#)
- [dbo:SoccerPlayer](#) [suy luận](#)
- [vio:Person Người](#) [suy luận](#)
- [vio:Athlete Vận động viên](#) [suy luận](#) [dbo:Athlete](#)
- [vio:FootballPlayer Cầu thủ bóng đá](#) [khai báo](#) [dbo:SoccerPlayer](#)
- [vio:NationalTeamPlayer Cầu thủ từng khoác áo đội tuyển quốc gia](#) [suy luận](#)

Liên kết dữ liệu mở

IRI: http://vi.dbpedia.org/resource/Nguyễn_Công_Phượng [Sao chép](#)

owl:sameAs: [dbr:Nguyễn_Công_Phượng · wd:Q18045362](#)

foaf:isPrimaryTopicOf: [vi.wikipedia.org/wiki/Nguyễn_Công_Phượng](#)

prov:wasDerivedFrom: [bản sửa đổi 75557409](#)

Tải RDF: [Turtle](#) · [N-Triples](#) · [JSON-LD](#) · [RDF/XML](#)

URI dereference được (303 + content negotiation):
`curl -L -H "Accept: text/turtle" http://127.0.0.1:8000/resource/Nguy%E1%BB%85n_C%C3%B4ng_Ph%C3%B2ng` [Sao chép](#)

SPARQL

Endpoint chuẩn SPARQL 1.1 Protocol: <http://127.0.0.1:8000/sparql> (JSON, XML, CSV theo header Accept; CONSTRUCT trả Turtle). Gọi từ dòng lệnh:

```
curl -G http://127.0.0.1:8000/sparql -H 'Accept: application/sparql-results+json' --data-urlencode 'query'
```

Chọn truy vấn mẫu... ☒ Có suy luận [Chạy \(Ctrl+Enter\)](#)

```
SELECT ?class (COUNT(DISTINCT ?x) AS ?n) WHERE {
  VALUES ?class { dbo:Person dbo:SoccerPlayer dbo:Organisation dbo:SoccerClub
  dbo:University
                    dbo:Place dbo:Province dbo:Stadium dbo:CareerStation }
  ?x a ?class .
}
GROUP BY ?class
ORDER BY DESC(?n)
```

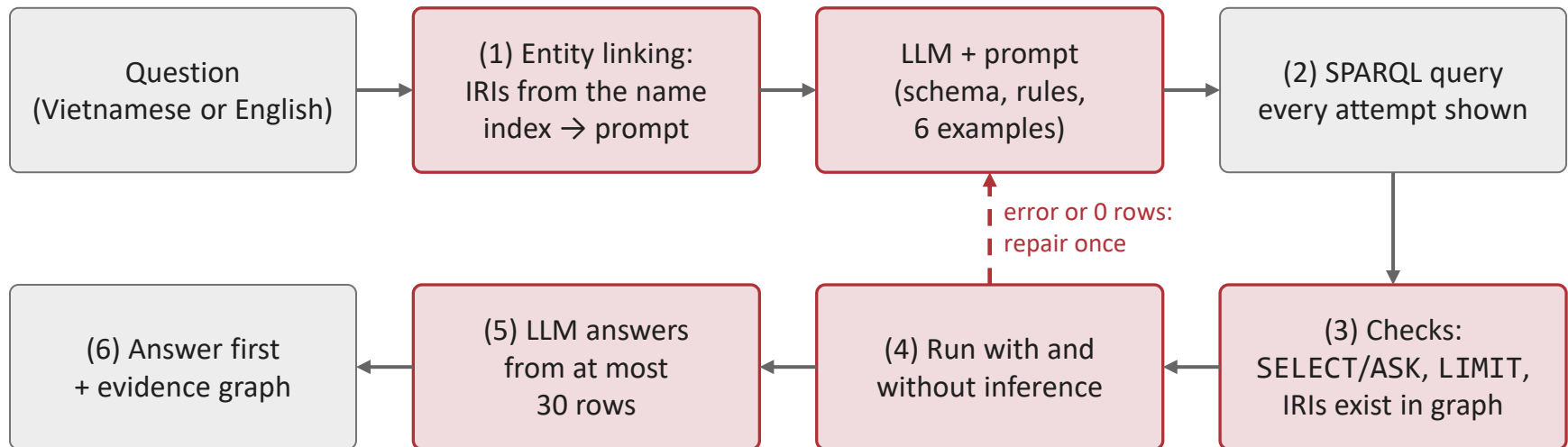
Các prefix vio, vres, dbo, rdfs, owl... đã khai báo sẵn. Truy vấn nặng không có LIMIT có thể chạy lâu.

SELECT · 9 dòng · 567 ms [có suy luận](#) [Tải JSON](#) [Tải CSV](#)

class	n
dbo:CareerStation	3382
dbo:Person	659
dbo:SoccerPlayer	611
dbo:Organisation	410
dbo:Place	370
dbo:SoccerClub	242
dbo:University	168
dbo:Province	110
dbo:Stadium	44

- **Overview, Ontology:** dataset size, 8,624 resources in the `vio:` class tree with its DBpedia parents, OWL axioms
- **Entity** (left): page like dbpedia.org/page/...; classes marked khai báo (asserted) or suy luận (inferred); Linked Data card, career timeline, draggable graph
- **Question answering:** Vietnamese or English, six visible steps (next slide)
- **SPARQL** (right): editor, 19 prefixes, 9 examples, inference switch, JSON/CSV
- **Hiện suy luận** switch: hides inferred triples in every graph

Question Answering: the LLM Writes SPARQL



- **Answer first, then six visible steps:** Than Quảng Ninh → 39 players with reasoning, 0 on the asserted data
- **Entity lookup by name in a subquery:** rdflib applies FILTER after the join, so the Công Phượng career query drops from **10.7 s** to **0.27 s**
- Works with any OpenAI-compatible API (gpt-4o in the demo, Gemini, Ollama) via `.env`; tests use a stubbed model, no API key needed

Evaluation: Dataset and Competency Questions

131,543

triples

990

entities

3,382

career stations

0

validation errors

88,898 asserted + 41,730 inferred + 859 ontology + 56 VoID

Question	Answer from the graph	Relies on
How many provinces are there now?	34 (76 former)	FormerProvince
Which unit succeeded Hà Tây?	Hà Nội	successor
Which teams has Công Phượng played for?	11 teams, 7 club stations	property chain
Which provinces have the most players?	Nghệ An 58, Hà Nội 48, Thanh Hóa 42	birthProvince
How many universities are in Hà Nội?	54	province
Who is “Công Phượng”?	Nguyễn Công Phượng	redirect

Every question is an automated test on the real data (lower bounds, so tests survive a re-crawl)

Limitations and Future Work

Limitations

- One domain, 990 entities — each new class needs a seed query, ontology terms, a mapping and a build function
- 44 of 777 DBpedia links are stale (12 redirects, 32 missing)
- 129 career stations without a team IRI; 26 universities without a province
- No time limit on `/sparql` queries: rdflib cannot cancel a running query
- Question answering not measured on a benchmark; its evidence graph is empty when rows hold one end of each relation

Future work

- Query time limits, or a triple store such as Virtuoso when the data grows; follow DBpedia redirects at build time
- More classes (coaches, competitions) and page links (`dbo:wikiPageWikiLink`)
- A benchmark of Vietnamese questions for question answering

Conclusion

A Vietnamese DBpedia for football, provinces and universities

Ontology `vio`: (18 classes, 54 properties) aligned with DBpedia; 990 Vietnamese articles → **131,543 triples**, 0 validation errors

Reasoning makes it interoperable

OWL 2 RL adds the DBpedia types and properties clients expect, and shortcut edges for multi-hop questions (`playedFor`: 0 → 2,772)

Five-star Linked Data with a query interface

777 links to English DBpedia (733 verified live), 990 to Wikidata; 303 + content negotiation, `/sparql` endpoint and terminal client, React UI with SPARQL editor and step-by-step question answering

Code and data: github.com/PhungHaThiKim/vietnamese-dbpedi

The background is a solid red color. Overlaid on this is a large, circular pattern of small white dots. The dots are arranged in concentric circles, with the density of the dots increasing towards the center, creating a halftone or dot-matrix effect. The pattern is centered on the slide.

Thank you
Questions?